

Continuous Architecture Assurance (CAA)

Calibrating AI reviewers as instruments — in one page

Rudson Carvalho · Independent Researcher

The thesis, in one line: CAA does not review your architecture. **CAA is the mechanism for measuring whether the things that review your architecture deserve to be trusted.**

The problem. Teams are starting to let AI review their architecture and code, and the reflex is to trust it — the model sounds competent, so its verdict is accepted. But an AI reviewer that has never been calibrated is not a reviewer; it is an opinion with a number on it. The market is full of "rulers" for judging software — well-architected frameworks, maturity models, checklists, AI reviewers — and nothing that tells you whether a ruler measures what it claims.

The idea. Treat an AI reviewer not as an oracle but as an **instrument to be calibrated**, the way a metrology lab calibrates a measuring device against a known standard. The standard is real incidents: does the reviewer catch the risk that actually caused a real outage (recall), and does it avoid crying wolf on a healthy system (false-positive rate)? The two numbers must be read as a pair — either alone is gameable. CAA is *ruler-agnostic*: your existing frameworks and reviewers plug in as instruments, provided they agree to be calibrated and to emit findings under a structured contract. The one rule that cannot be broken: the party that assures cannot be the party that produces.

Does it work? The honest scorecard. I built CAA and then spent five experimental attempts trying to *falsify* it, not confirm it. All numbers below are reconstructed from raw records:

Claim	Status
Contract-bound review reduces false positives vs. naive review	Demonstrated (two independent corpora)
The paired recall/FP metric catches degenerate reviewers	Demonstrated
Recall discriminates among <i>plausible</i> reviewers	Not demonstrated (recall saturates on textual cases)
The metric catches a confident-misattribution reviewer	Untested (the control resisted three construction attempts)
A closed-rule automated assurance gate is sufficient on its own	Refuted (it failed in all five runs; only human audit caught each)

That last row is the most important finding, and it arrived sideways: the automated gate I built to police the experiment accepted a serious defect *every single time*, and only an independent human reading the raw data caught it. For a framework whose entire pitch is "don't trust the reviewer, verify it," there is no more fitting result — the experiment proved its own founding rule.

Scope, stated plainly. This is a pilot. It establishes that a contract-bound reviewer can be calibrated against controlled *textual* incident pairs — not that CAA detects risk directly from live production systems, codebases, or telemetry. The experiment ran in Brazilian Portuguese (artifacts preserved in the original language); cross-language generalization is untested. The honesty about where it does *not* work is the reason to trust where it does.

When it is worth it. Use CAA where AI participates in reviewing or operating systems and being *confidently wrong* is expensive — regulated, financial, high-stakes domains. Skip it where the stakes don't justify the overhead. Minimum adoption: plug *one* reviewer under contract into a process you already have, make it emit a structured verdict with traceable evidence, and measure it against a handful of your own real incidents.

Go deeper

- **Understand the whole program** — the problem, the concepts, the full five-attempt saga, the verdict, and how to adopt it: [docs/technical-report.md](#)
- **The rigor** — three genre-separated papers (the framework, the pre-registered empirical study, the recursive gate-failure meta-paper): [docs/papers/](#)
- **The code, the corpus, the raw data** — reproducible, every number reconstructable independently of the harness: this repository.

A pilot falsification report, honestly scoped — not a finished product. Which is exactly what the framework would require me to say.

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